Supporting Information for

Predictive Modeling of COVID-19 Variant Peak Prevalence and Duration Using GISAID Data Across 15 Countries

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Overview. This appendix provides comprehensive supporting analyses that extend and contextualize the main results. It includes full country-level summaries of SARS-CoV-2 Pango lineage distributions across all 15 study countries, detailed definitions and visualizations of country-specific analytical periods, explicit variant inclusion and exclusion criteria across different early input windows, complete descriptions of predictive feature sets, and extended cross-country model performance evaluations. Together, these materials document the end-to-end data processing pipeline, justify key methodological and design choices, and demonstrate the robustness, stability, and generalizability of the proposed predictive framework across heterogeneous surveillance settings and sequencing environments.

1. Cross-country model performance and input-window comparison

Across the 15 countries evaluated, the model shows strong overall agreement between predicted and observed variant dominance**,** particularly for variants that ultimately reached both high peak prevalence (>20%) and long persistence (≥100 days)**.** In high-sequencing countries such as the United States, United Kingdom, Germany, Denmark, and Canada, most major variants—including BA.2, BA.5 sublineages, BQ.1.1, XBB.1.5, and JN.1—were correctly classified for both peak magnitude and duration. Errors were concentrated primarily among early pandemic lineages**,** where peak prevalence was often underestimated, reflecting sparse early data and rapidly changing surveillance practices. Duration predictions were generally more robust than peak predictions, even when peak magnitude was misclassified. Country-level sequencing volumes and variant composition underlying these comparisons are summarized in Table S1.

In contrast, greater heterogeneity appears in countries with noisier early trajectories or more abrupt epidemic transitions, such as South Korea, Japan, Brazil, and parts of Southern Europe. In these settings, the model frequently underpredicted peak prevalence—particularly for Delta-related AY lineages and some late Omicron sublineages—while still correctly identifying long-duration dominance. This pattern suggests that early explosive growth in some countries was difficult to distinguish from stochastic spikes using short early windows, whereas persistence signals were more stable. Overall, the table highlights a consistent pattern: peak prediction is more sensitive to early sampling variability and national context**,** while duration prediction remains accurate across diverse surveillance environments**,** supporting the model's utility for anticipating sustained epidemiological impact even when peak intensity is uncertain, particularly in heterogeneous surveillance settings.

Table S1. Comparison of observed and predicted variant dominance of all 15 countries. Dominance was defined as duration ≥100 days and peak prevalence >20%.

Country	Lineage	Onset date	Offset date	Pred. Peak	Obs. Peak	Peak. Diff	Pred. Dur.	Obs. Dur.	Dur. Acc.
United States	B.1.2	2020-03-09	2023-07-17	0.10–0.20	>0.20	F	31–100	100+	F
	Q (B.1.1.7)	2020-03-12	2024-03-04	0.10–0.20	>0.20	F	31–100	100+	F
	BA.2 (BA.2)	2020-04-03	2024-08-22	>0.20	>0.20	T	100+	100+	T
	XBB.1.5	2020-04-04	2024-07-16	>0.20	>0.20	T	100+	100+	T
	AY.103	2020-05-14	2023-02-01	>0.20	>0.20	T	100+	100+	T
	BG (BA.2.12.1)	2020-08-18	2024-08-28	>0.20	>0.20	T	100+	100+	T
	BF (BA.5.2.1)	2021-11-05	2024-04-11	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2021-11-30	2024-03-27	>0.20	>0.20	T	100+	100+	T
	HV.1	2023-01-09	2024-08-16	>0.20	>0.20	T	100+	100+	T
	JN.1	2023-01-09	2024-08-31	>0.20	>0.20	T	100+	100+	T
	JN.1.4	2023-09-13	2024-08-29	>0.20	>0.20	T	100+	100+	T
	others (n = 68)	2020-03-02	2024-08-16	-	-	-	-	-	-
United Kingdom	B.1.1	2020-03-02	2022-06-23	0.10–0.20	>0.20	F	31–100	100+	F
	Q (B.1.1.7)	2020-03-03	2022-03-03	0.10–0.20	>0.20	F	100+	100+	T
	BA.2 (BA.2)	2021-01-01	2024-01-13	>0.20	>0.20	T	100+	100+	T
	BA.5.1 (BA.5.1)	2022-01-12	2024-05-10	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2022-01-17	2024-02-12	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2022-02-11	2024-01-01	>0.20	>0.20	T	100+	100+	T
	XBB.1.5	2022-10-30	2024-05-28	0.10–0.20	>0.20	F	100+	100+	T
	XBB.1.16	2023-01-01	2024-04-17	>0.20	>0.20	T	100+	100+	T
	JN.1.4	2023-09-18	2024-08-25	>0.20	>0.20	T	100+	100+	T
	JN.1	2023-09-21	2024-08-24	>0.20	>0.20	T	100+	100+	T
	others (n = 76)	2020-03-02	2024-07-10	-	-	-	-	-	-
Germany	B.1.1	2020-03-09	2023-04-06	0.10–0.20	>0.20	F	31–100	100+	F

	A	2020-03-14	2022-03-04	0.10–0.20	>0.20	F	31–100	100+	F
	Q (B.1.1.7)	2020-03-15	2022-06-29	>0.20	>0.20	T	100+	100+	T
	AY.122	2020-05-11	2022-08-25	>0.20	>0.20	T	100+	100+	T
	AY.43	2020-05-11	2022-06-01	>0.20	>0.20	T	100+	100+	T
	BA.1.1 (BA.1.1)	2021-04-30	2023-05-23	>0.20	>0.20	T	100+	100+	T
	BA.2 (BA.2)	2021-04-30	2024-01-23	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2022-04-01	2023-04-06	>0.20	>0.20	T	100+	100+	T
	BA.5.1 (BA.5.1)	2022-04-07	2023-06-14	>0.20	>0.20	T	100+	100+	T
	BE.1.1	2022-04-07	2024-01-17	>0.20	>0.20	T	100+	100+	T
	BF.7	2022-06-13	2023-05-27	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2022-08-24	2023-07-10	>0.20	>0.20	T	100+	100+	T
	XBB.1.5	2022-11-04	2023-12-06	>0.20	>0.20	T	100+	100+	T
	EG.5.1	2023-03-30	2024-01-15	>0.20	>0.20	T	100+	100+	T
	others (n = 47)	2020-03-09	2024-01-01	-	-	-	-	-	-
Japan	B.1.1.214	2020-05-15	2021-07-29	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2020-08-15	2024-01-05	0.10–0.20	>0.20	F	31–100	100+	F
	BF (BA.5.2.1)	2020-08-15	2023-11-20	>0.20	>0.20	T	100+	100+	T
	BF.5	2020-08-15	2023-12-28	0.10–0.20	>0.20	F	100+	100+	T
	Q (B.1.1.7)	2020-10-02	2021-10-01	>0.20	>0.20	T	100+	100+	T
	R.1	2020-11-30	2021-09-14	0.10–0.20	>0.20	F	100+	100+	T
	BA.1.1.2 (BA.1.1.2)	2021-01-06	2023-04-13	0.10–0.20	>0.20	F	31–100	100+	F
	AY.29	2021-01-28	2023-11-06	>0.20	>0.20	T	100+	100+	T
	BA.2 (BA.2)	2021-12-30	2024-08-17	0.10–0.20	>0.20	F	31–100	100+	F
	BA.2.3.1 (BA.2.3.1)	2022-01-08	2023-05-25	>0.20	>0.20	T	100+	100+	T
	XBB.1.5	2022-12-01	2024-01-31	>0.20	>0.20	T	100+	100+	T
	HK.3	2023-01-01	2024-07-31	0.10–0.20	>0.20	F	31–100	100+	F
	XBB.1.16	2023-03-03	2024-04-01	>0.20	>0.20	T	100+	100+	T
	JN.1	2023-08-24	2024-08-17	0.10–0.20	>0.20	F	100+	100+	T
	others (n = 37)	2020-03-23	2024-08-03	-	-	-	-	-	-
Denmark	AY.43	2021-06-30	2022-02-28	>0.20	>0.20	T	31–100	100+	F
	BA.2.9 (BA.2.9)	2021-11-15	2022-11-20	>0.20	>0.20	T	100+	100+	T
	BA.2 (BA.2)	2021-11-22	2023-04-10	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2022-03-28	2023-02-22	>0.20	>0.20	T	100+	100+	T
	BA.5.1 (BA.5.1)	2022-05-03	2023-03-11	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2022-08-31	2023-04-09	>0.20	>0.20	T	100+	100+	T
	others (n = 49)	2021-06-21	2023-04-23	-	-	-	-	-	-
Canada	A	2020-03-16	2022-03-01	0.10–0.20	>0.20	F	31–100	100+	F
	B.1.1	2020-03-16	2023-12-21	0.10–0.20	>0.20	F	100+	100+	T
	B.1.2	2020-03-16	2021-10-07	0.10–0.20	>0.20	F	100+	100+	T
	B.1.128	2020-03-20	2021-01-13	0.10–0.20	>0.20	F	100+	100+	T
	Q (B.1.1.7)	2020-03-27	2021-11-15	>0.20	>0.20	T	100+	100+	T
	AE.8	2020-06-29	2021-06-17	>0.20	>0.20	T	100+	100+	T
	B.1.438.1	2020-09-24	2021-12-07	>0.20	>0.20	T	100+	100+	T
	AY.25.1	2020-11-18	2022-04-28	>0.20	>0.20	T	100+	100+	T
	P.1.14	2020-12-04	2021-09-18	>0.20	>0.20	T	100+	100+	T
	AY.27	2021-04-13	2022-04-28	>0.20	>0.20	T	100+	100+	T
	BA.1.1 (BA.1.1)	2021-10-29	2024-05-21	>0.20	>0.20	T	100+	100+	T
	BA.2 (BA.2)	2021-11-29	2024-07-09	>0.20	>0.20	T	100+	100+	T
	BA.2.3 (BA.2.3)	2021-12-27	2023-04-12	0.10–0.20	>0.20	F	31–100	100+	F
	BG (BA.2.12.1)	2022-01-24	2023-03-27	0.10–0.20	>0.20	F	100+	100+	T
	XBB.1.5	2022-03-20	2024-07-12	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2022-04-13	2024-08-15	>0.20	>0.20	T	100+	100+	T
	BF (BA.5.2.1)	2022-05-13	2024-03-16	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2022-08-30	2024-03-02	>0.20	>0.20	T	100+	100+	T
	EG.5.1	2022-10-07	2024-06-14	>0.20	>0.20	T	100+	100+	T
	HV.1	2023-07-31	2024-05-16	>0.20	>0.20	T	100+	100+	T
	JN.1	2023-10-09	2024-09-01	>0.20	>0.20	T	100+	100+	T
	JN.1.4	2023-10-16	2024-09-01	>0.20	>0.20	T	100+	100+	T
	others (n = 55)	2020-03-16	2024-08-26	-	-	-	-	-	-
France	B.1.1	2020-03-03	2023-02-14	0.10–0.20	>0.20	F	31–100	100+	F
	B.1.160	2020-03-13	2022-05-21	0.10–0.20	>0.20	F	31–100	100+	F

	Q (B.1.1.7)	2020-03-14	2022-02-09	>0.20	>0.20	T	100+	100+	T
	BA.2 (BA.2)	2020-03-28	2024-04-30	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2020-10-15	2024-05-16	0.10–0.20	>0.20	F	31–100	100+	F
	AY.43	2021-01-12	2022-07-21	>0.20	>0.20	T	100+	100+	T
	AY (B.1.617.2)	2021-04-06	2022-05-18	0.10–0.20	>0.20	F	100+	100+	T
	AY.122	2021-05-03	2022-09-09	>0.20	>0.20	T	100+	100+	T
	BA.5.1 (BA.5.1)	2021-05-17	2023-09-17	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2021-11-15	2023-09-08	0.10–0.20	>0.20	F	31–100	100+	F
	XBB.1.5	2022-02-20	2024-01-26	0.10–0.20	>0.20	F	100+	100+	T
	JN.1	2023-01-03	2024-08-10	>0.20	>0.20	T	100+	100+	T
	JN.1.1	2023-01-04	2024-08-10	0.10–0.20	>0.20	F	31–100	100+	F
	XBB.1.16	2023-02-05	2023-12-26	>0.20	>0.20	T	100+	100+	T
	others (n = 65)	2020-03-02	2024-07-02	-	-	-	-	-	-
Sweden	AY.122	2020-11-19	2022-03-21	0.10–0.20	>0.20	F	100+	100+	T
	BA.2 (BA.2)	2021-03-25	2024-04-01	>0.20	>0.20	T	100+	100+	T
	AY.4	2021-05-28	2022-02-07	>0.20	>0.20	T	100+	100+	T
	AY.43	2021-06-22	2022-02-21	>0.20	>0.20	T	31–100	100+	F
	BA.2.9 (BA.2.9)	2021-12-14	2023-12-12	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2022-01-11	2024-01-09	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2022-04-27	2024-05-20	>0.20	>0.20	T	100+	100+	T
	BA.5.1 (BA.5.1)	2022-05-04	2023-07-08	>0.20	>0.20	T	100+	100+	T
	XBB.1.5	2022-11-24	2024-05-05	0.10–0.20	>0.20	F	100+	100+	T
	JN.1	2023-01-02	2024-08-18	0.10–0.20	>0.20	F	100+	100+	T
	JN.1.4	2023-11-01	2024-08-16	>0.20	>0.20	T	100+	100+	T
	others (n = 67)	2020-03-23	2024-07-19	-	-	-	-	-	-
Spain	B.1.1	2020-03-02	2023-03-17	0.10–0.20	>0.20	F	100+	100+	T
	QB.1.177 (B.1.177)	2020-03-02	2022-05-11	>0.20	>0.20	T	100+	100+	T
	BA.1.17 (BA.1.17)	2020-11-06	2023-08-16	0.10–0.20	>0.20	F	31–100	100+	F
	AY.4	2020-11-13	2023-02-22	>0.20	>0.20	T	100+	100+	T
	AY.43	2020-11-20	2022-12-17	0.10–0.20	>0.20	F	100+	100+	T
	BA.2 (BA.2)	2021-02-15	2024-07-19	>0.20	>0.20	T	31–100	100+	F
	BA.2.9 (BA.2.9)	2021-07-14	2024-02-27	>0.20	>0.20	T	100+	100+	T
	BA.5.1 (BA.5.1)	2021-07-16	2024-06-11	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2021-07-16	2023-07-06	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2021-07-16	2024-03-25	0.10–0.20	>0.20	F	31–100	100+	F
	JN.1	2021-07-16	2024-08-09	0.10–0.20	>0.20	F	31–100	100+	F
	XBB.1.5	2021-07-16	2024-07-18	>0.20	>0.20	T	100+	100+	T
	others (n = 64)	2020-03-03	2024-07-17	-	-	-	-	-	-
Brazil	P.2	2020-06-26	2022-06-08	>0.20	>0.20	T	100+	100+	T
	P.1	2020-08-07	2022-07-25	>0.20	>0.20	T	100+	100+	T
	BA.1 (BA.1)	2020-08-15	2024-01-06	>0.20	>0.20	T	31–100	100+	F
	BA.1.1 (BA.1.1)	2021-01-12	2023-03-02	>0.20	>0.20	T	100+	100+	T
	AY.99.2	2021-02-28	2023-06-19	0.10–0.20	>0.20	F	100+	100+	T
	BA.2 (BA.2)	2021-11-17	2024-02-16	0.10–0.20	>0.20	F	100+	100+	T
	BA.5.1 (BA.5.1)	2022-01-06	2023-04-10	>0.20	>0.20	T	100+	100+	T
	BF (BA.5.2.1)	2022-01-06	2023-11-21	>0.20	>0.20	T	100+	100+	T
	BQ.1.1	2022-06-19	2023-06-03	>0.20	>0.20	T	100+	100+	T
	FE.1.2	2022-06-28	2023-11-27	0.10–0.20	>0.20	F	100+	100+	T
	FE (XBB.1.18.1)	2022-10-26	2023-09-21	0.10–0.20	>0.20	F	100+	100+	T
	XBB.1.5	2022-11-09	2024-01-26	0.10–0.20	>0.20	F	100+	100+	T
	GK.1	2023-03-24	2024-02-21	0.10–0.20	>0.20	F	100+	100+	T
	GK.1.1	2023-05-23	2024-02-06	0.10–0.20	>0.20	F	100+	100+	T
	JD.1.1	2023-06-27	2024-03-06	>0.20	>0.20	T	100+	100+	T
	others (n = 54)	2020-06-01	2024-03-10	-	-	-	-	-	-
Australia	A	2020-03-10	2022-03-22	0.10–0.20	>0.20	F	100+	100+	T
	Q (B.1.1.7)	2020-03-10	2021-08-16	0.10–0.20	>0.20	F	100+	100+	T
	D.2	2020-03-19	2020-11-17	>0.20	>0.20	T	100+	100+	T
	AY.39.1.2	2021-06-25	2022-06-14	>0.20	>0.20	T	100+	100+	T
	BA.2 (BA.2)	2021-12-01	2024-07-30	>0.20	>0.20	T	100+	100+	T
	BR.2.1	2022-01-11	2023-06-20	0.10–0.20	>0.20	F	100+	100+	T
	BA.5.2 (BA.5.2)	2022-01-18	2023-08-29	>0.20	>0.20	T	100+	100+	T

	BF (BA.5.2.1)	2022-02-18	2023-04-13	>0.20	>0.20	T	100+	100+	T
	XBF	2022-08-06	2023-10-02	>0.20	>0.20	T	100+	100+	T
	XBB.1.5	2022-10-22	2024-03-19	0.10–0.20	>0.20	F	100+	100+	T
	HK.3	2023-01-04	2024-05-03	>0.20	>0.20	T	100+	100+	T
	JN.1	2023-01-17	2024-08-16	>0.20	>0.20	T	100+	100+	T
	XBC.1.6	2023-01-17	2024-02-11	0.10–0.20	>0.20	F	31–100	100+	F
	XBB.1.16	2023-02-28	2024-01-06	>0.20	>0.20	T	100+	100+	T
	others (n = 58)	2020-03-09	2024-07-01	-	-	-	-	-	-
	B.1.619.1	2021-02-22	2021-09-21	>0.20	>0.20	T	100+	100+	T
	AY (B.1.617.2)	2021-04-22	2023-07-11	0.10–0.20	>0.20	F	100+	100+	T
	AY.69	2021-05-15	2022-02-10	0.10–0.20	>0.20	F	100+	100+	T
	AY.122.5	2021-08-08	2022-03-04	0.10–0.20	>0.20	F	100+	100+	T
	BA.2.3 (BA.2.3)	2021-12-29	2023-03-15	>0.20	>0.20	T	100+	100+	T
South Korea	BF (BA.5.2.1)	2022-05-10	2023-04-15	0.10–0.20	>0.20	F	31–100	100+	F
	BA.5.2 (BA.5.2)	2022-05-11	2024-01-22	>0.20	>0.20	T	100+	100+	T
	XBB.1.5	2022-11-23	2023-12-11	0.10–0.20	>0.20	F	100+	100+	T
	EG.5.1.1	2023-05-16	2024-03-27	0.10–0.20	>0.20	F	100+	100+	T
	HK.3	2023-07-04	2024-04-15	>0.20	>0.20	T	100+	100+	T
	JN.1	2023-10-24	2024-04-28	0.10–0.20	>0.20	F	31–100	100+	F
	others (n = 37)	2021-02-08	2024-02-20	-	-	-	-	-	-
	AE (B.1.1.306)	2020-04-27	2021-12-06	0.10–0.20	>0.20	F	31–100	100+	F
	B.1.1	2020-04-27	2023-05-31	0.10–0.20	>0.20	F	100+	100+	T
	B.1.36	2020-04-28	2022-07-09	0.10–0.20	>0.20	F	31–100	100+	F
	AY (B.1.617.2)	2020-05-01	2023-01-09	>0.20	>0.20	T	100+	100+	T
	B.1.36.29	2020-06-27	2021-11-01	>0.20	>0.20	T	31–100	100+	F
	BA.2 (BA.2)	2020-08-01	2024-01-10	>0.20	>0.20	T	100+	100+	T
	BA.2.75 (BA.2.75)	2020-08-08	2023-04-03	>0.20	>0.20	T	31–100	100+	F
India	BA.2.10 (BA.2.10)	2021-09-07	2023-04-19	>0.20	>0.20	T	100+	100+	T
	BA.2.38 (BA.2.38)	2022-01-07	2022-09-24	0.10–0.20	>0.20	F	100+	100+	T
	XBB.2	2022-01-08	2023-08-21	0.10–0.20	>0.20	F	100+	100+	T
	XBB	2022-05-30	2023-10-16	0.10–0.20	>0.20	F	100+	100+	T
	XBB.1	2022-09-04	2023-12-07	>0.20	>0.20	T	100+	100+	T
	XBB.2.3	2022-12-09	2024-01-04	>0.20	>0.20	T	100+	100+	T
	XBB.1.16	2023-01-04	2024-01-08	>0.20	>0.20	T	100+	100+	T
	others (n = 49)	2020-04-27	2024-01-11	-	-	-	-	-	-
	Q (B.1.1.7)	2020-03-06	2022-03-08	0.10–0.20	>0.20	F	100+	100+	T
	BA.5.1 (BA.5.1)	2020-07-20	2023-08-18	>0.20	>0.20	T	100+	100+	T
	BA.5.2 (BA.5.2)	2020-07-20	2023-12-13	0.10–0.20	>0.20	F	100+	100+	T
	BA.2 (BA.2)	2020-07-22	2024-01-20	0.10–0.20	>0.20	F	100+	100+	T
	AY.4	2021-01-03	2022-02-16	>0.20	>0.20	T	100+	100+	T
Italy	AY.43	2021-01-03	2022-03-16	>0.20	>0.20	T	100+	100+	T
	BA.1.1 (BA.1.1)	2021-01-03	2023-07-12	>0.20	>0.20	T	31–100	100+	F
	BQ.1.1	2022-01-07	2023-12-16	0.10–0.20	>0.20	F	100+	100+	T
	XBB.1.5	2022-02-19	2024-01-12	>0.20	>0.20	T	100+	100+	T
	XBB.1.16	2023-02-19	2024-01-18	>0.20	>0.20	T	100+	100+	T
	others (n = 59)	2020-02-24	2024-01-21	-	-	-	-	-	-
	AY (B.1.617.2)	2021-01-17	2022-10-31	>0.20	>0.20	T	31–100	100+	F
	AY.122	2021-04-26	2022-08-16	>0.20	>0.20	T	100+	100+	T
Austria	AY.43	2021-06-19	2022-08-16	>0.20	>0.20	T	100+	100+	T
	XBB.1.5	2022-08-01	2023-11-21	>0.20	>0.20	T	100+	100+	T
	others (n = 29)	2020-12-29	2023-12-01	-	-	-	-	-	-

For the prediction of weekly peak prevalence, the SuperLearner ensemble consistently outperformed all individual models across input windows. Using a 14-day input, SuperLearner achieved an accuracy of 0.59 and a macro-F1 of 0.53, improving to an accuracy of 0.76 and macro-F1 of 0.70 with a 21-day input, and maintaining similarly strong performance at 28 days (accuracy = 0.75, macro-F1 = 0.70). In contrast, the best-performing single models, such as CART and SVM, exhibited lower accuracy and greater variability across input windows. These results demonstrate the ensemble's ability to stably integrate complementary strengths of linear, nonlinear, and tree-based learners to capture early growth signals predictive of peak dominance.

Table S2 (a). Model performance for predicting the Weekly Peak Share

Model	Input of 14 days		Input of 21 days		Input of 28 days	
	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1
GLM	0.49	0.28	0.59	0.44	0.61	0.49
Bayes GLM	0.49	0.28	0.58	0.44	0.62	0.51
GAM	0.50	0.31	0.61	0.47	0.67	0.55
Elastic Net	0.48	0.35	0.57	0.45	0.57	0.41
SVM	0.53	0.43	0.65	0.59	0.66	0.61
Neural Net	0.48	0.64	0.55	0.48	0.65	0.58
CART	0.55	0.41	0.72	0.69	**0.77**	**0.72**
SuperLearner	**0.59**	**0.53**	**0.76**	**0.70**	0.75	0.70

Table S2 (b). Model performance for predicting the Time over 10% share.

Model	Input of 14 days		Input of 21 days		Input of 28 days	
	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1
GLM	0.40	0.32	0.45	0.40	0.44	0.38
Bayes GLM	0.40	0.32	0.45	0.41	0.44	0.39
GAM	0.42	0.35	0.47	0.41	0.46	0.42
Elastic Net	0.35	0.34	0.42	0.39	0.43	0.37
SVM	0.58	0.51	0.68	0.60	**0.70**	**0.67**
MLP	0.48	0.44	0.54	0.51	0.59	0.57
CART	0.48	0.42	0.53	0.49	0.56	0.53
SuperLearner	**0.61**	**0.57**	**0.70**	**0.72**	0.67	0.57

2. Descriptive data and analytical-period definition

Table S3 supplements Table 2 in the main text by extending the descriptive summaries to all 15 study countries. It reports year-by-year sequencing counts and the five most prevalent raw Pango lineages in each country, providing the full country-level context for the descriptive statistics discussed in the main analysis.

Table S3. Year-by-year distribution of raw Pango lineages and sequencing counts across the 15 study countries, reporting the total number of cases available for genomic analysis and the five most prevalent lineages in each year.

Country	Year	Number of Cases	Top Five PANGO Lineages (Annual Proportional Share)
United States	2020	18,903,169	B.1 (1.1%); B.1.1 (1.1%); B.1.595 (1.1%); B.1.369 (1.1%); B.1.243 (1%)
	2021	196,861,939	B.1 (1.0%); B.1.617.2 (1%); AY.44 (0.9%); B (0.9%); AY.25 (0.9%)
	2022	227,680,907	BA.2 (0.7%); BA.1.1 (0.6%); BA.2.12.1 (0.6%); BA.2.3 (0.6%); BA.2.9 (0.5%)
	2023	71,328,981	XBB.1.5 (0.5%); XBB.1.5.10 (0.4%); XBB.1 (0.4%); XBB.1.5.15 (0.4%); XBB.1.5.49 (0.4%)
	2024	9,457,427	JN.1 (1.1%); JN.1.8 (1.1%); JN.1.18 (1%); JN.1.7 (1%); JN.1.4 (1.0%)
United Kingdom	2020	13,728,198	B.1.1 (1.2%); B.1 (1.2%); B.1.1.37 (1%); B.1.1.1 (1%); B.1.177 (0.9%)
	2021	84,393,505	AY.4 (1.5%); B.1.617.2 (1.5%); AY.122 (1.5%); AY.5 (1.5%); AY.111 (1.5%)
	2022	79,974,012	BA.2 (1.3%); BA.2.3 (1.2%); BA.1.1 (1.2%); BA.2.10 (1.2%); BA.1 (1.2%)
	2023	8,416,483	XBB.1.5 (0.9%); XBB.1.9.1 (0.7%); CH.1.1.1 (0.7%); XBB.1.5.7 (0.7%); XBB.1.5.18 (0.7%)
	2024	934,727	JN.1 (2%); JN.1.7 (1.9%); KP.2 (1.7%); KP.3 (1.6%); JN.1.8 (1.5%)
Germany	2020	108,912	B.1.1 (7.3%); B.1 (6.5%); B.1.258 (3.7%); B.1.177 (3.6%); B.1.160 (3.4%)
	2021	20,390,838	B.1.617.2 (1.4%); AY.122 (1.3%); AY.4 (1.1%); AY.126 (1.1%); AY.9.2 (1.1%)
	2022	36,740,424	BA.2 (1.3%); BA.2.9 (1.1%); BA.2.3 (1%); BA.2.65 (1%); BA.2.10 (0.9%)
	2023	6,236,183	XBB.1.5 (0.7%); XBB.1 (0.7%); CH.1.1 (0.7%); BQ.1.1 (0.6%); XBB.1.5.46 (0.6%)
	2024	56,691	JN.1 (6.5%); KP.3.3 (3.6%); KP.3.1.1 (3.4%); JN.1.16 (3.3%); JN.1.4 (2.9%)
Japan	2020	134,325	B.1.1.284 (14%); B.1.1.214 (13.8%); B.1.1 (8.5%); B.1.560 (4.3%); B.1 (4%)
	2021	2,034,956	B.1.1.7 (7.8%); B.1.617.2 (6.8%); AY.29 (6%); AY.75.3 (5.3%); AY.29.1 (4.9%)
	2022	26,631,154	BA.2.3 (1%); BA.2 (0.9%); BA.5.2.1 (0.9%); BA.5.1 (0.9%); BA.5.2 (0.9%)
	2023	9,390,301	XBB.1.5 (0.8%); BF.7.15 (0.6%); BN.1.2 (0.6%); BN.1.2.2 (0.6%); BF.7.4.1 (0.6%)

	2024	646,792	XDQ.1 (2.3%); JN.1 (2.2%); XDQ (2.1%); JN.1.4 (2.1%); BA.2.86.1 (2%)
	2020	2,278,524	B.1.1 (1.5%); B.1 (1.5%); B.1.1.298 (1.5%); B.1.160 (1.5%); B.1.1.277 (1.4%)
	2021	11,304,092	B.1.617.2 (1.9%); AY.122 (1.7%); AY.4 (1.7%); AY.46 (1.6%); AY.43 (1.5%)
Denmark	2022	17,001,863	BA.2 (1.7%); BA.2.9 (1.6%); BA.2.3 (1.3%); BA.2.65 (1.3%); BA.1.1 (1.3%)
	2023	1,581,532	XBB.1.5 (1%); CH.1.1.12 (0.8%); CH.1.1 (0.8%); XBB.1 (0.8%); BQ.1.1 (0.7%)
	2024	77,590	JN.1.20 (2.6%); KP.2 (2.3%); KP.3.1 (2.3%); KP.3.3 (2.3%); KP.3 (2.3%)
	2020	584,489	B.1.1.176 (3.9%); B.1.1 (3.8%); B.1 (3.5%); B.1.279 (3.5%); B.1.128 (3.3%)
	2021	6,797,225	B.1.617.2 (2.4%); AY.27 (2.2%); AY.74 (2%); B.1.1.7 (2%); AY.122 (1.9%)
Canada	2022	12,176,039	BA.2 (1.1%); BA.2.3 (1%); BA.2.12.1 (1%); BA.5.1 (0.9%); BA.1.1 (0.9%)
	2023	8,189,558	XBB.1.5 (1.1%); XBB.1.16 (0.8%); EG.5.1.1 (0.7%); XBB.1.16.6 (0.6%); EG.5.1 (0.6%)
	2024	1,374,129	KP.2 (1.7%); KP.3 (1.7%); JN.1.4 (1.7%); JN.1 (1.7%); JN.1.7 (1.6%)
	2020	79,452	B.1 (8.7%); B.1.1 (6.6%); B.1.160 (6.4%); B.1.177 (6%); B.1.1.269 (4%)
	2021	12,230,918	B.1.617.2 (1.5%); AY.122 (1.4%); AY.43 (1.4%); AY.5 (1.4%); AY.4 (1.4%)
France	2022	22,865,501	BA.2 (1.1%); BA.2.9 (0.9%); BA.2.3 (0.8%); BA.1.18 (0.8%); BA.1.1 (0.8%)
	2023	5,633,065	XBB.1.5 (0.7%); XBB.1.9.1 (0.6%); XBB.1.16 (0.6%); XBB.2.3 (0.5%); XBB.1.17.1 (0.5%)
	2024	313,992	KP.3.1.1 (1.8%); KP.3.1 (1.7%); KP.2.2 (1.6%); KP.2.3 (1.6%); KP.3 (1.6%)
	2020	17,246	B.1.1 (12.7%); B.1 (11.4%); B.1.177 (5.4%); B.1.160 (5.3%); B.1.177.82 (4.4%)
	2021	3,920,346	B.1.1.7 (2.5%); B.1.617.2 (2.1%); B.1.351 (2.1%); AY.122 (1.8%); AY.4 (1.6%)
Sweden	2022	3,592,183	BA.2 (1.7%); BA.2.9 (1.5%); BA.2.65 (1.4%); BA.5.2.1 (1.2%); BA.5.1 (1.2%)
	2023	1,136,251	XBB.1.5 (1.3%); EG.5.1.1 (1.1%); EG.5.1.3 (1%); HK.3 (1%); JN.2 (1%)
	2024	51,081	JN.1.16 (3.8%); KP.3.1 (3.5%); JN.1 (3.1%); KP.3.1.1 (3.1%); JN.1.4 (2.7%)
	2020	89,756	B.1 (11.2%); B.1.177 (8.2%); B.1.1 (8.1%); A.2 (4.9%); A.5 (4.6%)
Spain	2021	2,937,437	B.1.617.2 (2.6%); B.1.1.7 (2.4%); AY.43 (2.3%); AY.42 (2.3%); AY.122 (2.2%)
	2022	3,185,817	BA.2 (2.5%); BA.2.9 (2%); BA.5.1 (1.7%); BA.2.3 (1.6%); BA.1.1 (1.5%)

Figure S1 illustrates the analytical periods defined in Stage 1 of the data workflow, with the study window indicated by two vertical blue lines. Within these boundaries, daily reported COVID-19 cases (black dots) are plotted to reflect the total denominators available for variant analysis. Across countries, the analytical periods are generally aligned to cover the major epidemic waves while excluding early sparse reporting and late low-case periods. For example, countries with sustained surveillance and larger case counts (e.g., United States, United Kingdom, Germany) have broader analytical windows, while countries with smaller denominators (e.g., Austria, Australia, South Korea) show narrower periods, ensuring that analysis is performed during intervals of reliable case volume.

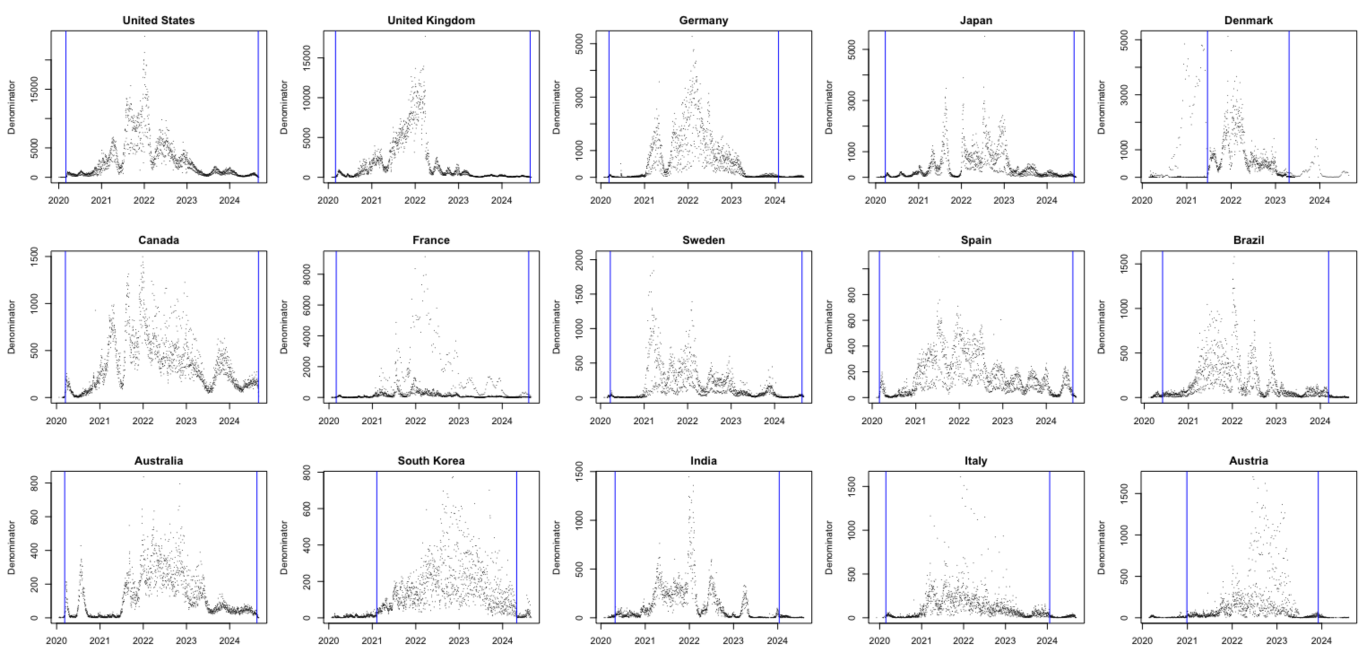

Figure S1: Analytical periods (blue vertical lines) defined for the 15 study countries, overlaid on their sequencing denominators across time

Figure S2 consists of three panels showing the major SARS-CoV-2 variants excluded from the analytical period for the United States (a), the United Kingdom (b), and South Korea (c). For each country, the six variants with the largest overall sample volumes were selected for display. As shown, these variants either circulated primarily before the onset of the analytical period (e.g., B.1, B.1.1) or after its end (e.g., KP.3.1.1, JN.1.16), and thus fell outside the defined analytical scope. This corresponds to the earlier description of the analytical period: the dashed blue vertical lines mark the study window, and variants outside this interval were excluded.

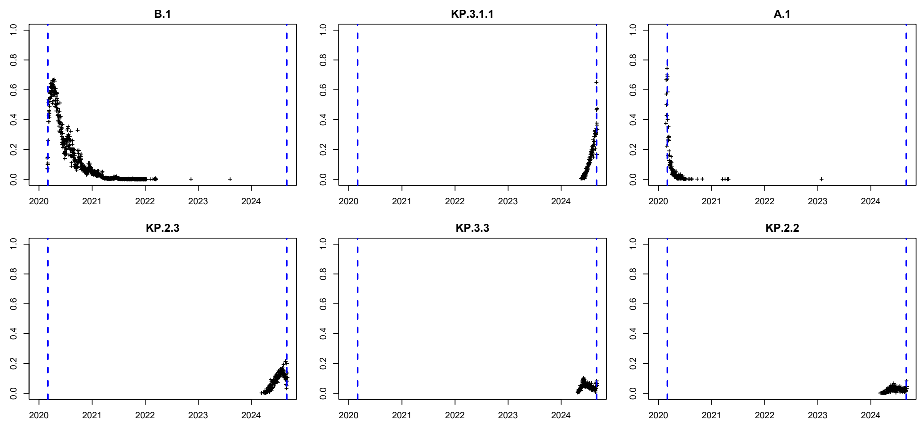

Figure S2 (a): Main SARS-CoV-2 variants from the United States in the GISAID dataset that were excluded from the analytical scope.

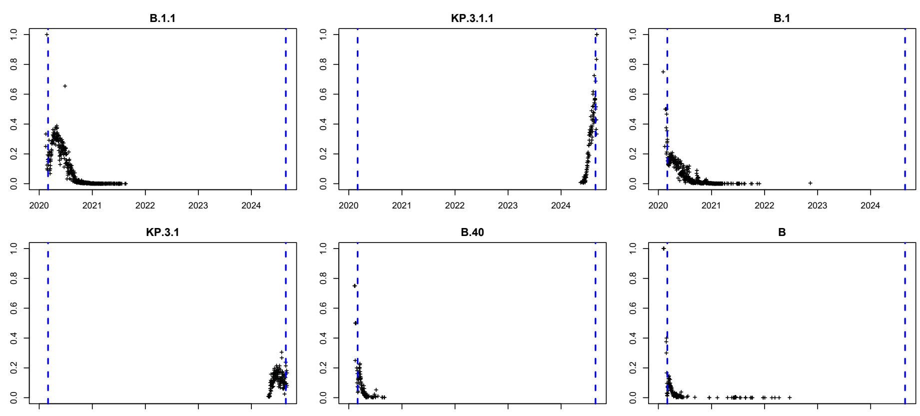

Figure S2 (b): Main SARS-CoV-2 variants from the United Kingdom in the GISAID dataset that were excluded from the analytical scope.

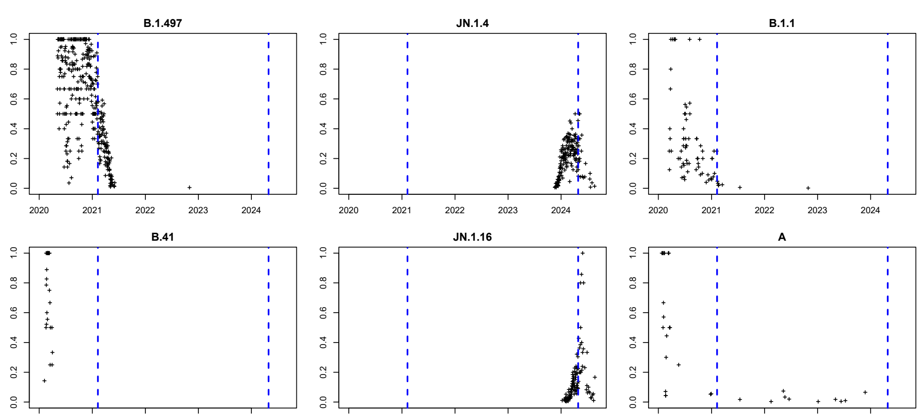

Figure S2 (c): Main SARS-CoV-2 variants from the South Korea in the GISAID dataset that were excluded from the analytical scope.

3. Predictive feature sets

Tables S4(a) and S4(b) summarize the feature sets used in the predictive models. Table S4(a) lists the six manually defined, epidemiologically motivated features that capture variant growth, prevalence, and sampling activity. Table S4(b) presents the *catch22* feature set, which provides statistically motivated descriptors of time-series dynamics, including distributional, autocorrelation, spectral, nonlinear, and motif-based properties. Together, they ensure a balance between domain-driven interpretability and statistical richness.

Table S4 (a). Manually defined epidemiological features used for modeling lineage time series.

#	Feature Name	Short Name	Category	Description
1	Log-linear slope	log_slope	Growth trend	Growth or decline estimated
2	Short-window slopes	short_slopes	Early growth	Slopes from first 7 days
3	Longest increasing run	inc_run	Temporal dynamics	Longest sequence of strictly increasing values
4	Slope difference	slope_diff	Growth trend comparison	Long-term minus short-window growth slope.

Table S4 (b). Statistically motivated catch22 features (plus extensions), grouped by conceptual category

#	Feature Name	Short Name	Category	Description
1	DN_HistogramMode_5	mode_5	Distribution shape	5-bin histogram mode
2	DN_HistogramMode_10	mode_10	Distribution shape	10-bin histogram mode
3	DN_OutlierInclude_p_001_mdrmd	outlier_timing_pos	Extreme event timing	Positive outlier timing
4	DN_OutlierInclude_n_001_mdrmd	outlier_timing_neg	Extreme event timing	Negative outlier timing
5	first1e_acf_tau	acf_timescale	Linear autocorrelation	First 1/e crossing of the ACF
6	firstMin_acf	acf_first_min	Linear autocorrelation	First minimum of the ACF
7	SP_Summaries_welch_rect_area_5_1	low_freq_power	Linear autocorrelation	Power in lowest 20% frequencies
8	SP_Summaries_welch_rect_centroid	centroid_freq	Linear autocorrelation	Centroid frequency
9	FC_LocalSimple_mean3_stderr	forecast_error	Simple forecasting	Error of 3-point rolling mean forecast
10	FC_LocalSimple_mean1_tauresrat	whiten_timescale	Incremental differences	Change in autocorr. timescale after differencing
11	MD_hrv_classic_pnn40	high_fluctuation	Incremental differences	Proportion of high incremental changes
12	SB_BinaryStats_mean_longstretch1	stretch_high	Symbolic	Longest stretch of above-mean values
13	SB_BinaryStats_diff_longstretch0	stretch_decreasing	Symbolic	Longest stretch of decreasing values
14	SB_MotifThree_quantile_hh	entropy_pairs	Symbolic	Entropy of successive pairs in symbolized series
15	CO_HistogramAMI_even_2_5	ami2	Nonlinear autocorrelation	Histogram-based AMI (lag 2, 5 bins)
16	CO_trev_1_num	trev	Nonlinear autocorrelation	Time reversibility
17	IN_AutoMutualInfoStats_40_gaussian_fmmi	ami_timescale	Linear autocorrelation	First minimum of AMI function
18	SB_TransitionMatrix_3ac_sumdiagcov	transition_variance	Symbolic	Transition matrix column variance
19	PD_PeriodicityWang_th001	periodicity	Linear autocorrelation	Wang's periodicity metric
20	CO_Embed2_Dist_tau_d_expfit_meandiff	embedding_dist	Linear autocorrelation	Exponential fit to embedding dist. distribution
21	SC_FluctAnal_2_rsrangefit_50_1_logi_prop_r1	rs_range	Self-affine scaling	Rescaled range fluctuation analysis (low-scale)
22	SC_FluctAnal_2_dfa_50_1_2_logi_prop_r1	dfa	Self-affine scaling	Detrended fluctuation analysis (low-scale)
23	DN_Mean	mean	Catch24 extension	Mean of raw series
24	DN_Spread_Std	std	Catch24 extension	Standard deviation of raw series

4. Variant exclusion by input-window criteria

Table S5 summarizes excluded variants across countries at different input windows (14–28 days), showing at which stage they failed to meet inclusion criteria. Variants were excluded either because they had already surpassed 30% share during the window or never reached 1%, with check marks indicating the reason for exclusion.

Table S5. List of excluded variants in different number of input days modeling and the reason to be excluded.

Country	Lineage	Exclusion Stage	Already Reached 30%	Never reach 1%
United States	AY.114	14 Days		✓
United States	AY.120	14 Days		✓
United States	AY.46.4	14 Days		✓
United States	XBB	14 Days		✓
United Kingdom	AY.127	14 Days		✓
United Kingdom	AY.36	14 Days		✓
United Kingdom	AY.4.8	14 Days		✓
United Kingdom	AY.9.2	14 Days		✓
Germany	B.1	14 Days	✓	
Germany	B.1.258	14 Days	✓	
Japan	P4 (B.1.1.284)	14 Days	✓	
Canada	B.1	14 Days	✓	
Canada	C57 (B.1.1.157)	14 Days	✓	
Sweden	B.1.258	14 Days	✓	
Sweden	B.1.351	14 Days	✓	
Brazil	P (B.1.1.28)	14 Days	✓	
Australia	AY.39.1	14 Days	✓	
Austria	Q (B.1.1.7)	14 Days	✓	
India	B.1	14 Days	✓	
Denmark	BA.1 (BA.1)	21 Days	✓	
Sweden	QB.1.177.82 (B.1.177.82)	21 Days	✓	
Australia	B.1	21 Days	✓	
Australia	BA.1.17 (BA.1.17)	21 Days	✓	
Brazil	BE.9	21 Days	✓	
Canada	B.1.279	28 Days	✓	
France	EG.5.1.3	28 Days	✓	
Italy	B.1	28 Days	✓	